

**To what extent do over-the-counter testosterone boosters and protein supplements influence hormonal balance, metabolic markers, and athletic performance in young athletes?**

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## **ABSTRACT**

This paper analyses the degree to which over-the-counter testosterone boosters and protein supplements affect hormonal homeostasis, markers of metabolism, and exercise performance among young athletes. As increasing dependence on nutritional supplements is used for performance gains, the investigation appraises adolescent athlete's physiological framework of hormonal control, immunity interactions, and growth-related processes. Testosterone boosters, promoted frequently by purported benefits of increasing strength, muscle, and libido, remain contentious owing to limited peer-reviewed research and questionable long-term safety. Protein supplements, especially those of plant origin like soy and pea protein, are assessed for muscle protein synthesis, recovery, and metabolic control, and their benefits and inconsistencies over age groups pointed out. By synthesizing information pertinent to endocrinology, immunity response, and muscle metabolism, the paper highlights the fact that even though there may be short-term performance benefits of some supplements, their efficacy and safety diverge greatly, posing the need for prudent use and additional empirical confirmation.

## **LITERATURE REVIEW**

The administration of dietary supplements for sports performance enhancement is a fast-evolving field of research in sports science, and of these, testosterone boosters (TBs) and protein supplements have been the most used substances. Published work available currently includes contradictory information regarding their usefulness, safety, and long-term consequences, especially among youth competitors. Testosterone plays an important role in controlling muscle mass, strength, and general sports performance. Although pharmacological testosterone replacement therapy is described in detail, over-the-counter TBs have a substantially less efficacious evidence base. TBs contain typically herbal

extractions, for example, *Tribulus terrestris*, *Eurycoma longifolia* (Tongkat Ali), fenugreek, and ashwagandha, and minerals, for example, zinc, boron, and magnesium.

Such studies as Clemesha et al. (2019) proved that almost 67% of TB components have no rigorous clinical validation, and few exhibit consistent serum testosterone-boosting activity. Some studies indicate performance gains whereby resistance exercise and TBs exhibit synergistic benefits, for example, enhanced muscle mass and  $\text{VO}_2$  max through ashwagandha supplementation. Other studies, however, show no appreciable alteration of testosterone levels, and here marketing-driven, rather than evidence-based, conclusions prevail. Most important, safety issues receive inadequate investigation, particularly prolonged use among teenagers, whose endocrine processes continue maturation development. Protein supplementation, and in particular after exercise, is widely researched to aid muscle protein synthesis (MPS).

Whey protein, casein, and colostrum from cows have been deemed highly effective at achieving net positive protein balance. Vegetable substitutes, including soy and pea protein, have been of growing popularity owing to their comparable amino acid compositions and digestions (NSCA, 2017). Soy protein was demonstrated to aid body composition and metabolic process in middle-aged men, although data on aged groups remain less trustworthy, and attenuated exercise-dependent strength gain was demonstrated in comparison with a dairy protein (Thomson et al., 2016). Pea protein, relatively less studied, showed promise in muscle damage prevention and recovery aid. Moreover, available literature points towards the mTOR activating effect of essential amino acid and insulin in stimulating MPS (Phillips, 2014). Currently, though, there is no strong indication of vegetable proteins achieving comparable long-duration performance gains of animal proteins.

Adolescence poses special physiological considerations in the assessment of supplementation. Growth, hormones, and energy needs differ substantially from adult groups, hindering the interpretation of supplement impacts. Research on energy deficiency in sport (RED-S) demonstrates an inadequate diet may compromise testosterone synthesis and immune capability among young athletes, frequently encouraging supplement use. Early sport specialization, differences in biological maturation, and hormonal differences, however, render it problematic to control for supplementation impacts. Early maturation, for example, may render males temporarily stronger and more resistant, and males, especially, may exhibit supplemental advantages in strength and endurance by virtue of higher supranatural levels of testosterone (Malina et al., 2004).

Although protein supplementation has been widely researched in terms of exercise performance, there is limited, and for the most part, inconclusive evidence supporting TBs. Longitudinal research on adolescent response to TBs is especially limited, generating questions of efficacy and safety. Not surprisingly, there is a parallel lack of comparative studies needed for plant-based proteins to define their relative efficacy against animal-based proteins in stimulating MPS, recovery, and exercise outcomes at varying ages. Broadly, there is a consensus in the literature for controlled, age-related trials necessary to derive sharper guidelines for young athletes' use of supplements.

## **INTRODUCTION**

In recent years, the use of dietary supplements among athletes, particularly young athletes, has grown at an unprecedented pace. Once considered niche products reserved for elite competitors, supplements have now become mainstream within the wider fitness culture. The global expansion of the sports nutrition industry, coupled with the influence of social media and the constant pressures of competition, has normalised supplement use as a

pathway to performance, appearance, and recovery. Young athletes today are increasingly turning to a wide variety of products, often marketed with promises of rapid muscle growth, improved endurance, or faster recovery. Among these, testosterone boosters and protein supplements stand out as two of the most widely promoted and consumed categories. Their popularity highlights both the aspirations of athletes to gain a competitive edge and the broader cultural emphasis on body image, strength, and achievement.

This growing reliance on supplements, however, raises critical questions about health, safety, and regulation. While athletes may seek to enhance their training outcomes, the potential consequences of improper use are often overlooked. There is limited awareness among many consumers—especially younger users—about the physiological risks, the gaps in regulatory oversight, and the possibility of misuse when products are obtained over the counter or online. Unlike pharmaceutical drugs, supplements are often regulated as food ingredients, which means they are not subject to the same strict requirements for safety, efficacy, or manufacturing standards. This creates an environment in which products may be contaminated, incorrectly labelled, or marketed with exaggerated claims. For athletes who are highly invested in performance outcomes, these uncertainties can have profound effects not only on health but also on the very goals they hope to achieve.

**Testosterone boosters** are a prime example of this dynamic. These products, often made from herbal extracts or marketed as precursors to testosterone, are promoted with the claim of naturally elevating hormone levels in order to stimulate muscle growth, increase strength, and enhance recovery. Such promises resonate strongly with athletes who are under pressure to maximise physical performance. Yet the benefits are often overstated, and the physiological risks are not always well understood by users. Prolonged or inappropriate use can disrupt the body's natural hormonal balance, placing strain on the liver, altering

metabolism, and contributing to long-term health issues. This tension—between the allure of rapid improvement and the hidden costs of misuse—captures why testosterone boosters have become a controversial supplement in sports nutrition.

**Protein supplements** form another widely used category. Whey, casein, and plant-based protein powders are promoted as tools to accelerate muscle repair, stimulate growth, and improve recovery following intense training. For athletes with demanding schedules, these products offer a convenient way to increase protein intake beyond what is easily achieved through diet alone. Protein supplements, therefore, occupy a central place in the routines of many young athletes, reinforcing their image as an essential part of training. Yet, as with testosterone boosters, the risks are not negligible. Overconsumption can place stress on the kidneys and liver, contribute to metabolic disruption, and may not even yield additional performance benefits once dietary protein needs are met. Athletes who adopt a “more is better” mentality risk undermining the very outcomes they seek to enhance.

Beyond these two categories, other supplements are also marketed with claims of supporting immunity or recovery. **β-glucans**, for example, are polysaccharides derived from yeast, fungi, algae, and oats, thought to stimulate the body’s innate immune system. While animal studies—such as those conducted in mice inoculated with influenza—have suggested potential benefits, research in human athletes has shown no significant effects. This illustrates a broader issue in sports supplementation: the gap between marketing claims and actual scientific evidence. Many products are consumed on the basis of perceived rather than proven benefits, creating a cycle in which athletes invest heavily in supplements that may not deliver the promised outcomes.

The issue extends further when considering the risks of misuse and contamination. Athletes often combine multiple supplements without accounting for total doses or potential

interactions, increasing the likelihood of side effects. Even widely used products such as caffeine and bicarbonate can cause nausea, gastrointestinal distress, anxiety, or sleep disturbances when consumed in excess. In more extreme cases, dangerous dosing errors have led to hospitalisation and even fatalities, as seen in incidents involving caffeine administration in laboratory settings. The lack of stringent regulation exacerbates these dangers. Because supplements are not monitored with the same rigour as pharmaceuticals, products have been found to contain toxic levels of vitamins, harmful contaminants such as lead or broken glass, or unlisted ingredients that pose serious health risks. While some of these issues resemble broader food safety problems, their occurrence during critical training or competition periods can have a disproportionate impact on athletes.

For these reasons, the widespread and growing use of supplements among young athletes cannot be viewed as a simple matter of personal choice or training preference. It represents a complex intersection of ambition, marketing, regulation, and health. Testosterone boosters and protein supplements exemplify both the appeal and the risks of this culture. They are marketed as accessible, almost essential, tools for success, yet they carry hidden costs that many users are unaware of.

In the rapidly developing world, diet and nutrition have attracted more and more attention, especially in sports-related fields used for promoting health and achieving the best body composition. Suitable control of diet and intake of supplements can efficiently advance sport-specific ability, avoid illness, and decrease the proportion of fat in the body. Protein is a key macronutrient for human health, but protein intake's effectiveness on exercise performance, and more specifically, plant protein intake continues to be questionable and disputable. Soya chunks is of a high quality protein and is a complete protein according to the National Strength and Conditioning Association's (NSCA) practitioner's guide to sport and

exercise nutrition. Its Protein Efficiency Ratio (PER) and Protein Digestibility-Corrected Amino Acid Score (PDCAAS) are equal to those of dietary meat or fish and slightly lower than egg, milk, casein, bovine colostrum, and whey protein, respectively, and thus intake of soy protein supplements is a realistic option for individuals.

On a more personal level, this topic resonates deeply with me as a national-level gymnast. Over the course of my training and competitions, I have been told on multiple occasions by coaches, peers, or others in the sporting community that I should consider taking supplements. These suggestions were often made casually and without reference to proper research or medical guidance. At times, I found myself questioning whether such products were truly necessary for progress or whether they might carry unseen risks. What struck me most was how little information was readily available to young athletes like myself about how these substances actually affect the body in the short and long term. *This personal experience highlighted for me the importance of asking critical questions about supplements:*

**Are the claimed benefits backed by evidence? Do the risks outweigh the potential gains?**

**And how can athletes make informed choices when the information available is often incomplete or misleading?**

Furthermore, the impact of plant proteins on young and old groups of persons proved heterogeneous. Whereas protein consumption from soya, alongside resistance exercise, augmented body composition and metabolic performance in middle-aged men, there was no significant difference in muscle performance and metabolic health in old groups of persons, according to other research. Thomson et al. indicated that increased intake of soy protein inhibited muscle strength gains across resistance exercise in old groups of persons in comparison with participants consuming dairy protein or control protein intake. Despite these contradictory views, pea protein is presented in the form of a promising supplement for



preserving muscle protein synthesis and exercise performance, and hence requires further research on its relative effectiveness in comparison with animal proteins. Pea protein proved beneficial in reducing muscle damage and muscle recovery.

## **PHYSIOLOGICAL FRAMEWORK**

### **Adolescent/young athlete physiology**

In mature societies, children's competitive sport is typically initiated at approximately 6–7 years of age. Accordingly, pre-teens may possess several years' exposure to intensive preparation and of elite standard international competition. Early selection into, and persistence in, elite sport takes place in a matrix of biocultural variables, of which the most significant are socioeconomic status, family size, family encouragement, health status, and psychological preparation. There exist, however, several physical and physiological variables affecting young adults' performance and persistence in sport, and these exert sport-specific operations. Bodyweight is of relevance, and elite young children's sports stars, like their adult counterparts, possess bodily types appropriate for their sport. In most sports, men and women possess, on average, median population values or superior heights, and in certain team games, heights vary according to self-selected position.

Gymnastics and figure skating are the only sports in which young competitors are regularly below population median sizes. Bodyweight acts similarly to, though less strongly than, height, and young competitors in most sports tend, on average, to have higher-than-average bodyweights and larger weights-to-height ratios than others of their age, though, again, gymnasts and figure skaters typically have low weights with weights-to-height ratios no larger than those of non-athletes. Young competitors continue to be thinner than non-athletes, and the difference is larger in girls than in boys. Chronological age-grouped sport advantages the boy maturing early and, thus, the person born in the earlier part of the

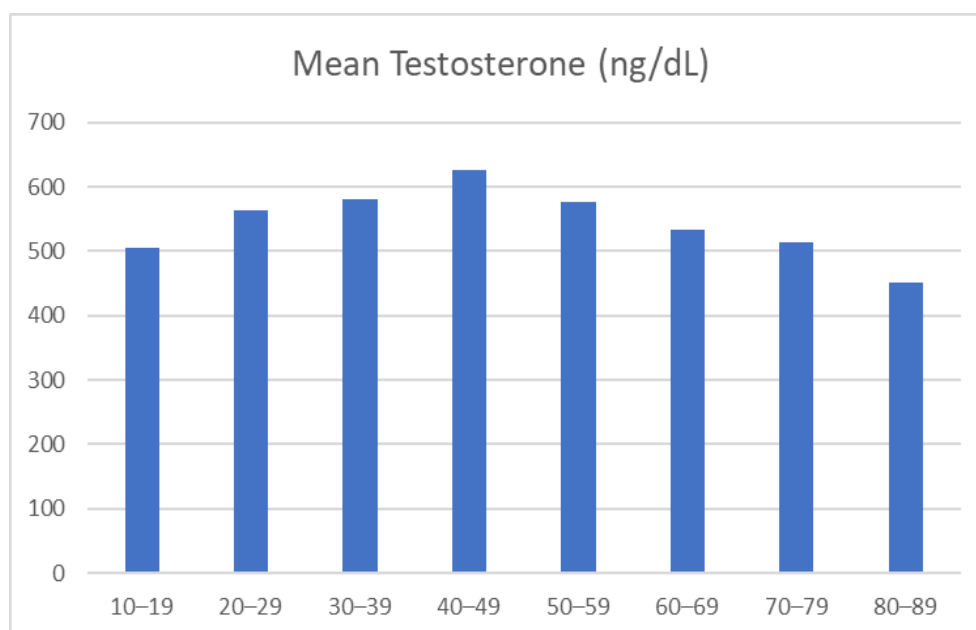
selection year, since he will tend to be biologically older and is, in fact, older than the boy born at the end of the selection year. Such a profile of success and failure has been observed in football, tennis, ice-hockey, and swimming. Perhaps the most clear example comes from a study of the birthdates of footballers selected for the School of Excellence of the Football Association, School of Excellence, UK.

In 6 years, roughly 16 14-year-old males were chosen each year for the School of Excellence. Birth distribution of the 103 males chosen showed 67% were born in the first quarter of the selection year and under 2% in the last 3 months of selection year. 89% of entrants were born in the first half of the selection year. Uneven birth distribution continues through senior games, whereby nearly half English premier league soccer players were born in the first third of youth selection year. Existing bias exhibited by senior team at end of selection year is almost certain to be due to selection at younger age; successful young children, by virtue of commanding more coaching, will be known and then chosen for senior team after selection at earlier age than unsuccessful children who will have attracted no selectors' attention and thus will have dropped back into regional ranks of soccer participation. Most gifted young children are denied selection and thus give up elite youth soccer simply by virtue of late birth in the selection year.

Mature girls of their age are heavier and taller having higher ratios of weight-to-height than their counterparts, and so have a head start in sport reliant on body size. Early maturers, on the other hand, have wide hips and shorter legs, and, in combination with puberty-associated increase in fatness of the body, these characteristics, under some sports, are of potential disadvantage. Early maturing girls do, of course, have an adolescent spurt of muscle mass, but it's limited compared with males. They will tend to be stronger than late-maturing girls, and aerobic and anaerobic power and fitness rise with maturation. However, physiological differences associated with maturation are smaller in girls than in males, and by

the age of 13–14 years, essentially eliminated. Late maturing girls have linear body types, reduced weight-for-height, low fatness, relatively longer legs, and smaller diameters of hip-shoulder, and these tend to be better adapted for success in sport. Early maturers, therefore, do not dominate women's sport.

Reported later menarche among girls in a number of sports, but few reliable data exist. Study sample sizes tend to be small and findings biased by both retrospective determination of the appearance of menarche and selective dropping of participants from some sports. Comparisons of menarche age distribution between athletes and non-athletes find more late-maturing than early-maturing girls, but most remain within the typical range of variation of menarche age. Moreover, the distance between menarche and peak height velocity age does not significantly differ between girls who participate in sports and girls who do not engage in sports. Only gymnasts regularly exhibit their menarche age averages more than two standard deviations from the population norm, but either deliberate selection of girls who mature late or dropping of girls who mature early from the programme could explain this finding.



### **Immune system interactions**

Currently, there is a vast array of supplements which, by producers, "elevate serum testosterone level"—supplements of this kind are termed "testosterone boosters" (TB). Generally, TB consists of a mix of a large array of natural compounds but does not include testosterone or other anabolic substances. Consumption of TB can be prevalent among sportsmen and women owing to grasping the role of testosterone in diverse components of sports performance, particularly strength, and with the continually enhanced identification of anabolic substances, which appear on the WADA Prohibited List. Another factor of the popularity of TB is the rising awareness of RED-S (Relative energy deficiency in sport) syndrome, including hypogonadal levels of testosterone among sportsmen and women, and the persistent search for lawful methods of elevating testosterone among such sportsmen and women.

At the same time, there is no evidence, both regarding safety and true effectiveness of TB as substances of elevating testosterone levels in the organism. The bulk of research on TB is concerned with treatment of erectile dysfunction, reduced libido, late-onset hypogonadism, and benign prostatic hyperplasia. Some of the components of TB might be efficient in the treatment of these conditions. Of course, there is increasing interest in the muscle strength and power-improving role of TB. One of the first studies in this area was conducted in Bulgaria, where among weightlifters, the use of *Tribulus terrestris* was actively promoted. Other substances are studied actively, too. Such, for instance, was the study of Chen et al., demonstrating that the use of a combination of *Eurycoma longifolia* and resistance training enhanced peak power output in young men. Lean body-mass change-boron intake correlation was addressed in the other study. Ashwagandha increases muscle mass] velocity, power, and maximal oxygen uptake (VO<sub>2</sub> max) in healthy young men, according to a new study. It is

believed that if TB does raise testosterone levels, then presumably, these might enhance strength, power, and some other testosterone-associated parameters, too.

### **Hormonal regulation of male/female reproductive hormones**

The regulation of the reproductive system by hormones is the activity of pituitary, adrenocortical, and gonadal hormones. During puberty, in males and females, the hypothalamus secretes gonadotrophin-releasing hormone (GnRH) so that the anterior pituitary will produce and release follicle stimulating hormone (FSH) and luteinizing hormone (LH) will be produced and secreted. These hormones control the gonads (testes in men and ovaries in women); these hormones are gonadotropins. FSH in males and females stimulates the production of gametes and LH stimulates the production of gonadal hormones. Elevated levels of gonad hormones suppress the production of GnRH through a negative feedback mechanism.

At the pituitary, GnRH stimulates the synthesis and release of the gonadotropins, FSH and LH. These processes are controlled by the pulse size and rhythm of GnRH and by feedback from androgens and estrogens. Low-frequency pulses of GnRH elicit the release of FSH, and high-frequency pulses of GnRH elicit the release of LH. FSH stimulates males' first division of meiotic spermatocytes, through the first division of meiosis, transforming secondary spermatocytes into mature cell sperm. FSH stimulates the production of androgen-binding protein by the testicular Sertoli cells, by acting on FSH receptors on their basolateral cell membranes. FSH synthesis is inhibited by the hormone inhibin, secreted and produced by the testes.

It stimulates the interstitial cells of the testes to secrete sex hormones (androgens). It is at times referred to as interstitial-cell-stimulating hormone. Testosterone, the best-known androgen of males, stimulates the development of the sperm and male qualities. A tiny

quantity of testosterone precursor is additionally produced in the adrenal cortex, although the function of this ancillary production of hormones is unknown. In women, FSH promotes maturation of egg cells (or ova) in glands known as follicles. Follicle cells, in turn, make the hormone inhibin, which suppresses production of FSH in the female reproductive tract. LH promotes development of ova, stimulates the process of ovulation, and promotes production of estradiol and progesterone by the ovaries. Estradiol and progesterone are steroid hormones and ready the female's body for bearing young. Estradiol forms female secondary sex characteristics, and both estradiol and progesterone control the menstrual cycle.

Aside from the synthesis of FSH and LH, the anterior section of the pituitary synthesizes the hormone prolactin (PRL) in both women and men. It promotes the synthesis of milk of the mammary glands after birth. Hypothalamic hormones, prolactin-releasing hormone (PRH) and prolactin-inhibiting hormone (PIH) (now known as dopamine), control prolactin. PRH stimulates and PIH suppresses the release of prolactin. The posterior pituitary secretes the oxytocin hormone, which acts on smooth muscles of the uterus to initiate contractions at birth. Smooth muscles of the uterus are oxytocin-insensitive early in pregnancy, but the oxytocin receptor population in the uterus increases late in pregnancy. Stretch of tissue of the cervix and uterus stimulates oxytocin release at birth. Contractions become stronger as oxytocin blood levels rise through a positive feedback loop until birth is finalized. Oxytocin also stimulates contraction of myoepithelial cells around secretory mammary glands, which form milk. By contraction, milk is forced from secretory alveoli into ducts and forced from breasts in a milk ejection ("letdown") reflex in response to suckling of an infant. Release of oxytocin is stimulated by suckling of an infant, evoking synthesis of oxytocin in the hypothalamus and release of oxytocin into the circulatory system at the posterior pituitary.

## **Protein synthesis of muscle and metabolic control**

The synergistic interactions between amino acid provision and resistance exercise on skeletal muscle protein synthesis rates (MPS) are now excellently explained. Dietary amino acid intake after resistance exercise promotes a rise in MPS and is critical in transitioning net protein balance (as  $MPS - \text{muscle protein breakdown (MPB)}$ ) from negative (net protein loss) to positive (net protein gain). Feeding-induced MPS changes in healthy individuals are ~3–5 times larger over any given day than measurable MPS changes in MPB, confirming MPS is extremely responsive, regulated, and the overwhelming controller of muscle net protein balance changes.

Inasmuch, then, for net muscle protein balance increases of sustained durations to accrue muscle mass, MPS would be extremely pertinent. We here are not positing, of course, that MPB is a biologically inconsequential process; MPB assists intramuscular amino acid content, and presumably plays a role in maintaining muscle protein quality through protein damage clearing and utilizing protein constituent amino acid components for the synthesis of new functional muscle protein molecules. Therefore, we posit nutrition interventions elevating MPS will be of significant scientific and clinical use as a tool of deriving favorable muscle protein balance and subsequent muscle protein accrual. Further, these interventions may be of use to exercise scientists examining ways of optimizing skeletal muscle's adaptive response under conditions of chronic exercise training.

Dietary amino acids and insulin are important nutrient-regulated effectors of MPS and MPB and there is recent work that has explained molecular pathways of the regulation of the rise in MPS by amino acids and contraction. MTORC1 protein kinase serves as a principal integration site of a wide range of favorable stimuli for MPS, such as dietary amino acids and contraction, at the molecular level. MTORC1 regulates MPS at the level of phosphorylation of downstream protein effectors of translation initiation of MPS, specifically, p70S6k and 4E-

BP1. Additionally, inhibition of mTOR activity by the drug rapamycin reduces the contraction, and EAA mediated increase in human MPS, and thus describes the important role of this mTORC1 kinase in MPS regulation.

Some studies suggest that amino acid supplementation after resistance exercise and consequent elevation of MPS was matched by increased phosphorylation of elements of the MTOR signaling cascade beyond levels after exercise alone without nutrition. Nevertheless, dissociation between direct MPS rate measurement and magnitude of muscle anabolic signaling molecule phosphorylation have been shown earlier. Moreover, precisely how amino acids come to stimulate mTORC1 to upregulate MPS is uncertain, though MAP4K3, vacuolar sorting protein 34 (Vps34), and Rag GTPases are signaling molecules in candidate amino acid 'sensor' roles within reach of sensing elevation of amino acid concentration and then activating mTORC1.

Furthermore, mRNA expression of several of these skeletal muscle amino acid transporters, i.e., LAT1 (SLC7A5), SNAT2 (SLC38A2), CD98 (SLC3A2), and PAT1 (SLC36A1) have been shown to be up-regulated after EAA ingestion and resistance exercise in human skeletal muscle. Such transporters may potentially participate in the regulation of human muscle protein metabolism through the capability of these transporters to bring amino acids beyond the cell membrane, and transmit information through downstream targets proposed to control MPS. Protein content of several of these amino acid transporters have also been shown to be up-regulated after EAA ingestion and resistance exercise, but it is not known at this time whether mRNA and protein expression increase corresponds to increased amino acid transport capacity.



## Neuroendocrine aspects

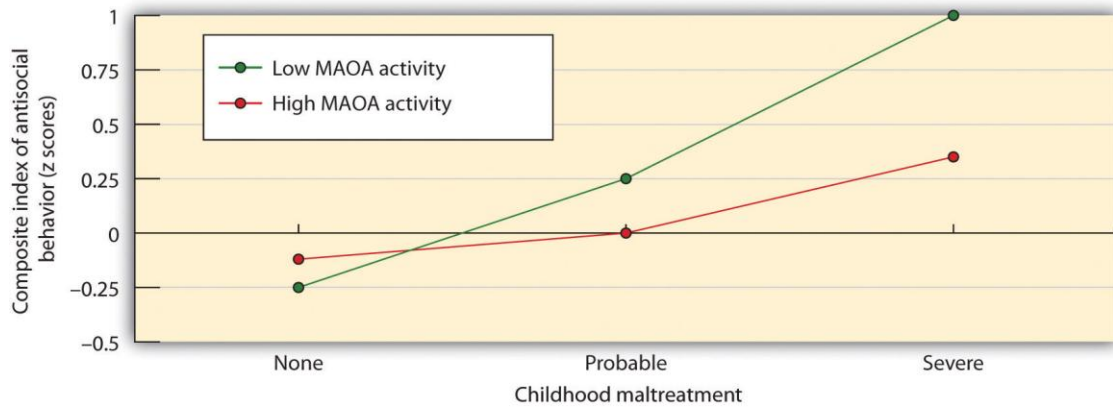
A belief in the inherent aggression of human beings that the capacity to aggress against others at least in some situations is part of our basic human nature is in accord with the principles of evolutionary psychology. After all, the self-preservative goal of maintaining and building the self - will in certain instances mandate that we deter others from doing us harm as well as those people for whom we care. Because it enables us access to valuable resources like foodstuff as well as preferred mates, or because it will safeguard us directly from being attacked by others, we may aggress toward others. Because our social ranking may also pose certain threats for us, and when our social ranking may threaten our survival prospects as well as the survival prospects of those people for whom we care so much, then too the process of natural selectivity may bring it about that human beings, as it would any other animal species, will aggress. In some circumstances human beings must be capable of being aggressive, and nature has equipped us for the purpose. In the proper circumstance, nearly all of us will aggress.

The effect of a specified genetic influence, the monoamine oxidase (MAOA) gene on the X chromosome, generates an enzyme involved in the production of serotonin, a chemical messenger involved in mood, appetite, and sleep as well as suppressing aggressiveness. In support of the influence of genetics on aggression itself, they determined that adults who had fewer activities of the gene as children were at higher risk for displaying an array of aggressiveness. Nonetheless, they also determined that the genetic influence was only significant for children who had also been maltreated highly.

While much more work is required, it seems as if aggression, as well as most other behaviors, is determined by an interplay between genes and the environment. Aggression is regulated in large measure by the region in the older part of the brain termed the amygdala.

The amygdala is an area of the brain involved in the regulation of our perceptions of, and responses to, fear and aggression. The amygdala connects with some other body systems involved in fear processes, including the sympathetic nervous system, facial movements, the processing of odors, and the release of the brain chemicals termed neuro-transmitters involved in stress and aggression. Apart from helping us feel fear, the amygdala also assists us in learning about situations that evoke fear. In reacting to positive events but also to negative ones, especially to stimuli by which we perceive something as threatening and fear evoking, when we encounter events that pose a danger, the amygdala acts on the brain so as to help it memorize the specifics of the situation so as to learn how not to encounter it in the future. The amygdala also gets activated when viewing facial expressions of others feeling fear or when confronted with members of racial outgroups.

In the past few years, researchers (and drug companies) have concentrated on the impact of deficiency in testosterone, primarily among males. Indeed, as males grow older, their levels of testosterone decrease very slowly, 1% to 2% per year as opposed to the comparatively fast decrease in estrogen resulting in the occurrence of menopause. The testes secrete less testosterone, there are fewer messages sent by the pituitary instructing the testes to secrete testosterone. Additionally, as males age, their livers synthesize more sex hormone binding globulin (SHBG), attaching itself to testosterone in the bloodstream. Hormone production in males is regulated by messages sent by the brain down through the pituitary gland at the base of the brain. The pituitary gland in turn sends messages to the testes to secrete testosterone. There's also a "feedback loop" tightly regulating the level of hormone in the blood. If the level of testosterone gets too high, messages go down through the brain to the pituitary gland instructing it to cut back production.



Source: BC Open Textbooks

## CONCLUSION

The analysis of available literature demonstrates the complicated and frequently inconclusive role of over-the-counter testosterone boosters and protein additives in determining hormonal equilibrium, metabolic indices, and athletic capability among young athletes. Although testosterone boosters are promoted on a large scale under the promise of increasing muscle mass, strength, and sexual desire, related scientific data remain scarce, unpredictable, and frequently found on small-scale, and non-replicable research. Additionally, the safety of long-term use, especially in adolescence when the endocrine apparatus continues its maturation, is not sufficiently clear.

Conversely, protein supplementation, and especially protein supplementation in the form of whey, casein, or other animal proteins, has a stronger evidence base in terms of aiding muscle protein synthesis, recovery, and general performance outcomes. Plant protein sources such as soy and pea protein offer legitimate substitutes, although their quality in comparison to animal proteins continues to be debated. Of particular value, protein supplementation is efficacious under most conditions of resistance training and proper dietary balance, thus supporting the argument that supplementation cannot replace planned nutrition and training regimes.

In youth participants, adolescent physiological alteration, variation in maturational levels, energy requirements, and sport-specific requirements make it difficult to evaluate the value of supplements. Accurate use of supplements, outside of nutrition, rest, and training principles, may expose the athlete to unnecessary health hazards while gaining minimal benefits in terms of performance. Overall, the evidence is that protein supplements can have an ancillary role in sport development, but testosterone boosters need rigorous analysis before use can be promoted safely, and so future research requires vigorous, age-matched, longitudinal trials, controlling for differential development, so young sportspersons' use of these agents is informed by evidence, not marketing rhetoric.

## REFERENCES

- <https://www.medicalnewstoday.com/articles/testosterone-booster>
- <https://www.webmd.com/healthy-aging/the-best-testosterone-boosters-for-men-over-50>
- <https://www.mdpi.com/2072-6643/16/16/2748>
- [https://bio.libretexts.org/Bookshelves/Introductory\\_and\\_General\\_Biology/General\\_Biology\\_\(Boundless\)/37%3A\\_The\\_Endocrine\\_System/37.07%3A\\_Regulation\\_of\\_Body\\_Processes\\_-\\_Hormonal\\_Regulation\\_of\\_the\\_Reproductive\\_System](https://bio.libretexts.org/Bookshelves/Introductory_and_General_Biology/General_Biology_(Boundless)/37%3A_The_Endocrine_System/37.07%3A_Regulation_of_Body_Processes_-_Hormonal_Regulation_of_the_Reproductive_System)
- <https://nutritionandmetabolism.biomedcentral.com/articles/10.1186/1743-7075-9-40>
- [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(05\)67845-2/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(05)67845-2/fulltext)
- <https://www.mdpi.com/2673-396X/2/2/11>
- <https://opentextbc.ca/socialpsychology/chapter/the-biological-and-emotional-causes-of-aggression/>
- <https://www.health.harvard.edu/staying-healthy/testosterone--what-it-does-and-doesnt-do>